FINM3093 Investments

Project Instruction

In the lecture, we are given the recipe of constructing one optimal risky portfolio using 2 risky assets. When multiple risky assets are included, we can locate the optimal risky portfolio by constructing the **efficient frontier**.

Basic project task: choose 5 stocks that interest you most, download the historical price/return data (ranging from Jan 2022 to Jan 2024) and construct the efficient frontier. Find the optimal risky portfolio and optimal complete portfolio. Backtest your strategy on recent half-year data (ranging from Feb 2024 to Aug 2024). Does your strategy perform well (say, compared with the market index)? How to improve?

Advanced task: estimates of mean and covariance can be important in determining the optimal portfolios. What change does it make to the optimal portfolios if you choose the dataset ranging from Jan 2020 to Jan 2022? What is the reason behind the changes in the portfolio composition? How to make a more robust portfolio decision?

Due date of report: Dec 12, 2024

3-4 students are recommended for each group. Each group is required to submit a report (no more than 15 pages for everything including the appropriate reference) on their own. You should explain the results in a way that is easy-to-read, and interpret your findings clearly. A 6-8 minutes' presentation is tentatively scheduled on **Dec 12**.

Evaluation methods: evaluation by the instructor (50%) + peer evaluation (50%)